

Simulation results

massive-binary-sim v0.1.0 · git `3b6e4a4840` · seed 42 · wallclock 0.34 s

1. Initial conditions

quantity	value
M1, M2 (ZAMS)	35, 25 Msun
separation / period	50 R _{sun} / 5.287 d
eccentricity	0.05
metallicity Z	0.0142
spins ($\Omega/\Omega_{\text{crit}}$)	0, 0

Active prescriptions

toggle	choice
gravity	`2.5PN`
wind	`vink2001`
convection	`ledoux`
overshoot	`exponential`
am_transport	`taylor_spruit`
mt_stability	`zeta`
mt_rate	`kolb_ritter`
ce_formalism	`alpha_lambda`
opacity	`kramers`
reaction_rates	`reaclib`
sn_engine	`delayed`
tides	`hut1981`
structure_fidelity	`semi_analytic`
third_body	`False`

2. Evolutionary timeline

t [Myr]	event	detail
0.016	star 1 -> MS	reduced-model phase
2.600	star 1 -> stripped_He	reduced-model phase
2.852	star 1 -> preSN	reduced-model phase
2.863	star 1 -> remnant	reduced-model phase
0.016	star 2 -> MS	reduced-model phase
7.533	star 2 -> HG	reduced-model phase
7.674	star 2 -> CHeB	reduced-model phase
7.674	star 2 -> stripped_He	reduced-model phase
8.007	star 2 -> preSN	reduced-model phase
8.007	star 2 -> remnant	reduced-model phase
2.599	**mass transfer_end** Case A transfer complete; donor -> 14.70 Msun stripped He star	
(at SN)	**SNI**	star 1 -> BH 13.23 Msun (delayed); bound after SN
7.674	**common_envelope** survived: a _f =9.034 R _{sun} (spiral-in x5.2)	
(at SN)	**SN2**	star 2 -> BH 5.65 Msun (delayed); bound after SN
8.007	**dco_formed** double compact binary (BH+BH) merging via GW in 326.0 Myr	

Fate: double compact binary (BH+BH) merging via GW in 326.0 Myr

3. Final-state table

quantity	star 1	star 2
mass [Msun]	13.23	5.653
phase	remnant	remnant
remnant	BH 13.23 Msun (delayed)	

```
| remnant (2) | | BH 5.653 Msun |
| separation | 8.064 Rsun | |
| period | 0.6105 d | |
| eccentricity | 0.2916 | |
```

4. Derived quantities

```
| quantity | value | unit | formula |
|---|---|---|---|
| chirp mass  $M_c$  | 7.397 | Msun |  $M_c = (m_1 m_2)^{3/5} / (m_1+m_2)^{1/5}$  |
| mass ratio  $q$  | 0.4273 | - |  $q = m_2 / m_1$  ( $\leq 1$ ) |
| symmetric mass ratio  $\eta$  | 0.2097 | - |  $\eta = m_1 m_2 / (m_1+m_2)^2$  |
| integrated wind loss (star1 / star2) | 0.3863 | Msun |  $\int \dot{M}_{\text{wind},1} dt$  ; star2 = 3.15 Msun |
| integrated RLOF transfer | 23.47 | Msun |  $\int \dot{M}_{\text{dot\_RLOF}} dt$  (gross, donor side) |
| net Delta  $M_1$  | 21.77 | Msun |  $M_1(\text{ZAMS}) - M_1(\text{final})$  |
| net Delta  $M_2$  | -19.35 | Msun |  $M_2(\text{final}) - M_2(\text{ZAMS})$  |
| orbital separation (final) | 8.064 | Rsun | `state variable` |
| orbital period (final) | 0.6105 | day |  $P = 2 \pi \sqrt{a^3 / G(m_1+m_2)}$  |
| orbital binding energy | -1.7587e+49 | erg |  $E = -G m_1 m_2 / (2a)$  |
| orbital angular momentum | 2.8244e+53 | g cm2 / s |  $J = \mu \sqrt{G M a (1-e^2)}$  |
| GW merger time | 3.2603e+08 | yr | `Peters 1964 eq. 5.14 (eccentric)` |
| GW merger time | 0.326 | Gyr | `as above` |
| peak GW frequency (ISCO) | 232.9 | Hz |  $f = c^3 / (6^{3/2} \pi G M)$  |
| char. strain at ISCO (D=100 Mpc) | 3.3357e-21 | - | `Maggiore 2007 eq. 4.44` |
```

5. Error budget

Numerical error (from the convergence study):

- final a converges to 3.000069 Rsun; rel error at $\text{rtol}=1e-8$ is 1.75e-06, at $1e-11$ is 1.07e-09; mean observed order ~ 1.07 per tol-decade
- relative error in final a vs the tightest tolerance: 1.7505e-06 at $\text{rtol}=1e-8 \rightarrow 1.0713e-09$ at $\text{rtol}=1e-11$
- mean observed order of convergence: 1.07 per tolerance decade (DOP853, 8th-order adaptive)
- **The numerical error ($\sim 1e-9$ relative) is far below the physical/model uncertainty quantified in sections 6-7.**

Physical / model uncertainty dominates and is quantified in §6-7. The reduced single-star evolution model (fitting formulae, not a structure)

6. Free-parameter sensitivity (one-at-a-time)

Outcome metric: **chirp_mass**. Parameters ranked by outcome spread across their literature range.

```
| rank | parameter | range | outcome spread | provenance |
|---|---|---|---|---|
| 1 | `beta_accretion` | 0-1 | 3.05 | fraction retained; poorly constrained, see Petrovic+ 2005 |
| 2 | `eta_wind` | 0.1-3 | 0.5548 | overall mass-loss multiplier (convention) |
| 3 | `clumping` | 1-10 | 8.3445e-04 | wind clumping factor; Vink 2001 rates assume smooth |
| 4 | `alpha_mlt` | 1.5-2.2 | 0 | solar-model calibrated; e.g. Magic+ 2015 A&A 573 A89 |
| 5 | `f_overshoot` | 0-0.04 | 0 | Brott+ 2011 A&A 530 A115; Claret & Torres 2018 ApJ 859 100 |
| 6 | `f0_overshoot` | 0-0.01 | 0 | Herwig 2000 A&A 360 952 (exponential-diffusive) |
| 7 | `alpha_sc` | 0.001-100 | 0 | Langer+ 1983 A&A 126 207 -- essentially unconstrained |
| 8 | `gamma_amlow` | 1-4 | 0 | specific AM of non-conservative outflow; Nelemans+ 2000 A&A 360 1011 |
| 9 | `alpha_ce` | 0.1-10 | 0 | CE efficiency; Ivanova+ 2013 A&ARv 21 59 |
| 10 | `lambda_ce` | 0.05-2 | 0 | envelope structure param; Xu & Li 2010 ApJ 716 114; Dewi & Tauris 2000 |
| 11 | `c12_ag_rate` | 0.6-1.6 | 0 | multiplier on  $12C(a,g)16O$  S-factor; deBoer+ 2017 RvMP 89 035007 |
| 12 | `kick_sigma` | 0-400 | 0 | Hobbs+ 2005 MNRAS 360 974 (km/s, Maxwellian 1D sigma) |
| 13 | `overshoot_step` | 0-0.5 | 0 | step overshoot in  $H_p$ ; Schaller+ 1992 / Ekstrom+ 2012 |
```

Dominant uncertain input: ``beta_accretion`` (outcome varies by 3.05 across its plausible range).

7. Theory comparison (competing prescriptions)

```
| toggle | option | default? | fate | M1 | M2 | Mchirp | a [Rsun] | log10 t_merge |
|---|---|---|---|---|---|---|---|---|
```

wind	`vink2001`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
wind	`bjorklund2021`		common-envelope merger (Case A, q=12.12)	2.86	34.68	n/a	58.13	n/a
convection	`schwarzschild`		double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
convection	`ledoux`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
overshoot	`none`		double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
overshoot	`step`		double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
overshoot	`exponential`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
am_transport	`none`		double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
am_transport	`tayler_spruit`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
mt_stability	`qcrit`		double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
mt_stability	`zeta`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
mt_rate	`ritter`		double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
mt_rate	`kolb_ritter`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
ce_formalism	`alpha_lambda`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
ce_formalism	`gamma`		wide double compact binary (BH+BH), GW merger	13.23	5.653	7.397	27.97	10.21
opacity	`electron_scattering`		double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
opacity	`kramers`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
reaction_rates	`nacre`		double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
reaction_rates	`reaclib`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
sn_engine	`rapid`		double compact binary (BH+BH) merging via GW i	13.23	7.596	8.661	9.414	8.748
sn_engine	`delayed`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
sn_engine	`direct_collapse`		double compact binary (BH+BH) merging via GW i	13.23	7.596	8.661	9.414	8.748
gravity	`newtonian`		double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
gravity	`2.5PN`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513
tides	`none`		common-envelope merger (Case B, q=1.37)	33.52	24.53	n/a	51.48	n/a
tides	`hut1981`	YES	double compact binary (BH+BH) merging via GW i	13.23	5.653	7.397	8.064	8.513

The scientific payload: the outcome is a mapped dependence on prescription choice, not a single number.

8. Validation suite results

test	status	detail
kepler_closure	PASS	200 orbits: max dE/E =1.77e-11, max dL/L =8.76e-12, da/a =1.52e-11 (tol 1e-06)
lane_emden_analytic	PASS	n=0: max dtheta =5.72e-16, dxil =8.88e-16; n=1: max dtheta =1.55e-14, dxil =1.82e-14; n=5: max dtheta =5.44e-14
periastron_advance_1PN	PASS	1PN d(omega)/orbit measured=6.157175e-05 rad, predicted 6 pi GM/(c^2 a(1-e^2))=6.155112e-05, frac diff=0.034%
hulse_taylor_decay	PASS	dPb/dt(Peters)=-2.402102e-12 s/s vs GR prediction -2.402630e-12 (Weisberg & Huang 2016); residual 0.022%. Merger
energy_conservation_2PN	PASS	2PN conservative: secular dE/E over 120 orbits = 1.68e-08 (tol 1e-07); max dL/L =1.12e-05
eddington_limit_flag	PASS	120 Msun ZAMS Gamma_Edd=0.406; 10x-luminosity state flags Gamma>1: True
solar_structure_eddington	PASS	Eddington model 1 Msun: P_c=1.256e+17 (ref 2.4e17), T_c=1.185e+07 (ref 1.57e7), rho_c=76.9 (ref 150), L=0.

9. Assumptions & limitations

- ****1-D, orbit-averaged.**** No 3-D hydrodynamics; the orbit is treated by secular equations except in the dedicated Kepler-closure test.
- ****Reduced stellar evolution.**** Single-star tracks are fitting formulae (Hurley+ 2000-style), not Henyey structure integrations. Core masses, radii and lifetimes carry ~10-30% systematic uncertainty.
- ****No magnetic fields**** beyond the Tayler-Spruit AM-transport toggle.
- ****No rotational mixing feedback**** on the tracks (rotation only sets Omega_crit and tidal rates here).
- ****Mass-transfer rate**** uses a Ritter/Kolb-Ritter-style analytic nozzle with an efficiency prefactor (KNOWN_GAPS BIN-MDOT).
- ****GW waveform**** is a 0PN Newtonian chirp -- qualitative, not for matched filtering.
- ****3.5PN dynamics**** not implemented (KNOWN_GAPS PN-35).
- Full list: KNOWN_GAPS.md.

10. Figures

- ****hr_diagram**** - `outputs/hr\_diagram.png`
- ****orbital_evolution**** - `outputs/orbital\_evolution.png`

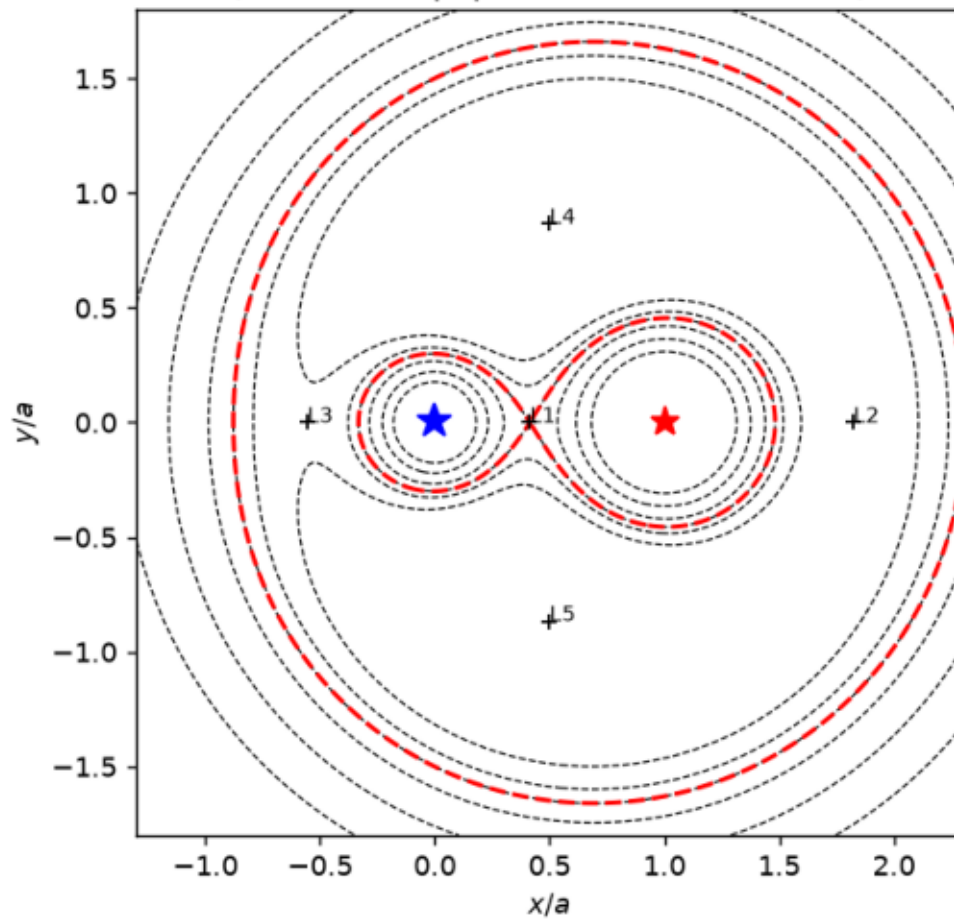
```
- **mass_evolution** - `outputs/mass_evolution.png`  
- **roche** - `outputs/roche.png`  
- **kippenhahn1** - `outputs/kippenhahn1.png`  
- **kippenhahn2** - `outputs/kippenhahn2.png`  
- **gw** - `outputs/gw.png`  
- **diagnostics** - `outputs/diagnostics.png`  
- **orbit_frame** - `outputs/orbit_frame.png`
```

11. Bibliography

See `REFERENCES.bib` for the full list with DOIs. Key prescriptions used in this run: Peters (1964); Mora & Will (2004); Eggleton (1983); Ritte

roche

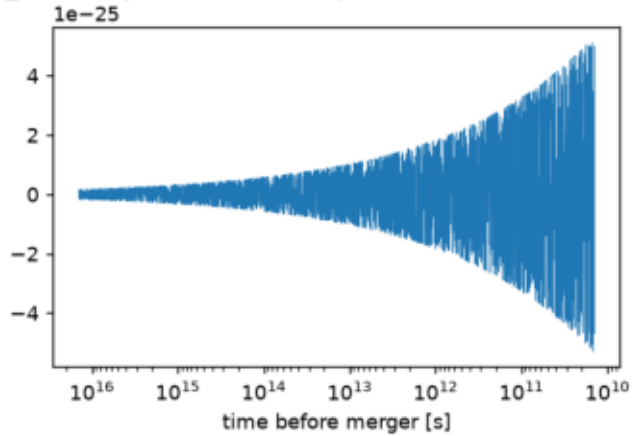
Roche geometry, $q=2.340$ (final)
(red = L1 equipotential / Roche lobes)



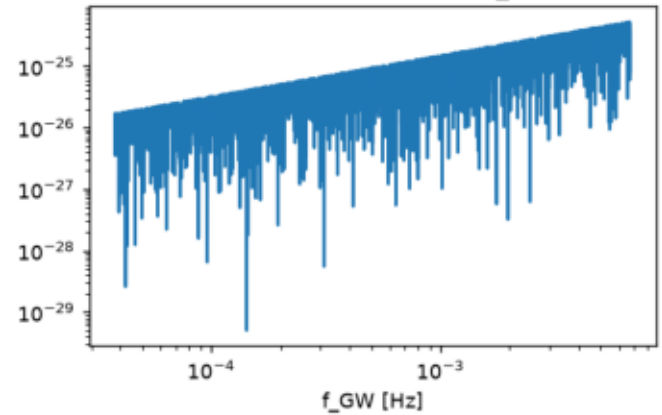
gw

Gravitational-wave output (gravity=2.5PN)

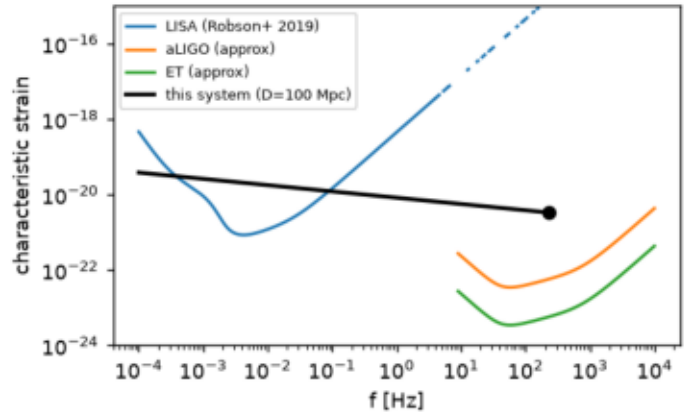
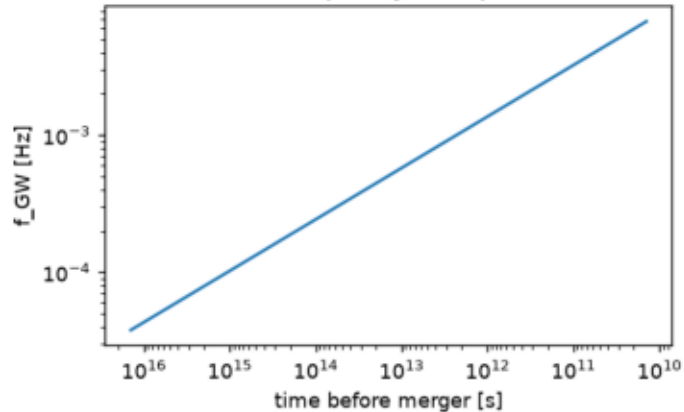
$h_+(t)$ chirp (0PN, $D=100$ Mpc); coalescence in 451 Myr



instantaneous strain vs f_{GW}



frequency sweep



orbit_frame

orbit @ t=1.81e+06 yr, a=38.3 R_{sun}, stars to scale

